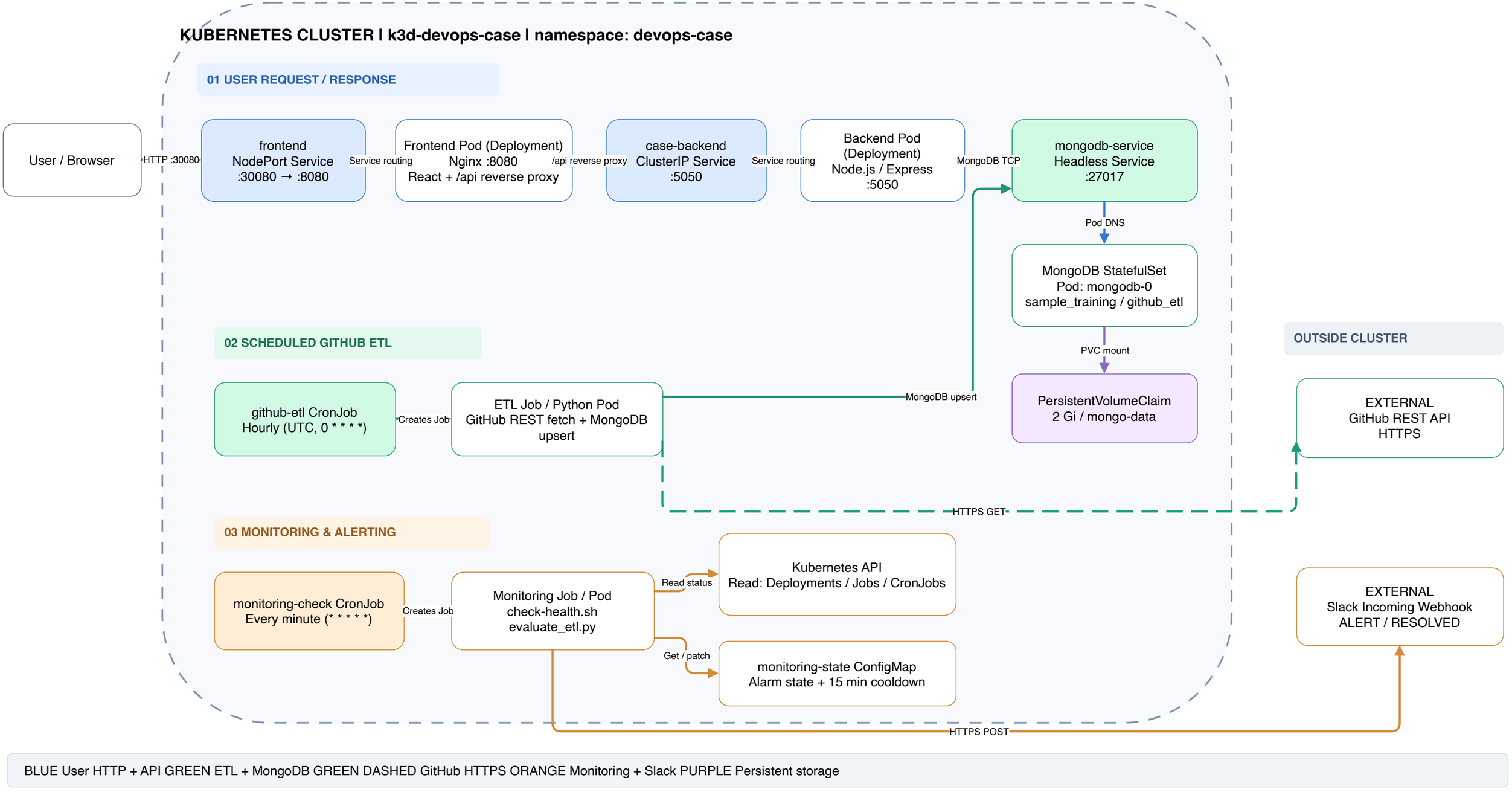

KUBERNETES CLUSTER | k3d-devops-case | namespace: devops-case
01 USER REQUEST / RESPONSE
User / Browser
HTTP :30080
frontend
NodePort Service
:30080 → :8080
Service routing
Frontend Pod (Deployment)
Nginx :8080
React + /api reverse proxy
/api reverse proxy
case-backend
ClusterIP Service
:5050
Service routing
Backend Pod
(Deployment)
Node.js / Express
:5050
MongoDB TCP
mongodb-service
Headless Service
:27017
Pod DNS
MongoDB StatefulSet
Pod: mongodb-0
sample_training / github_etl
PVC mount
PersistentVolumeClaim
2 Gi / mongo-data
02 SCHEDULED GITHUB ETL
github-etl CronJob
Hourly (UTC, 0 * * * *)
Creates Job
ETL Job / Python Pod
GitHub REST fetch + MongoDB
upsert
MongoDB upsert
OUTSIDE CLUSTER
EXTERNAL
GitHub REST API
HTTPS
HTTPS GET
03 MONITORING & ALERTING
monitoring-check CronJob
Every minute (* * * * *)
Creates Job
Monitoring Job / Pod
check-health.sh
evaluate_etl.py
Read status
Kubernetes API
Read: Deployments / Jobs / CronJobs
Get / patch
monitoring-state ConfigMap
Alarm state + 15 min cooldown
EXTERNAL
Slack Incoming Webhook
ALERT / RESOLVED
HTTPS POST
BLUE User HTTP + API GREEN ETL + MongoDB GREEN DASHED GitHub HTTPS ORANGE Monitoring + Slack PURPLE Persistent storage